

Let A be an $n \times n$ complex matrix, and define its numerical range by

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n \text{ and } \|x\|_2 = 1\},$$

where x^* denotes the conjugate transpose of x . Then, for every complex polynomial p ,

$$\|p(A)\|_2 \leq 2 \max_{z \in W(A)} |p(z)|,$$

where $\|p(A)\|_2$ is the operator norm induced by the Euclidean norm. Equivalently, the closure of the numerical range $W(A)$ is a 2-spectral set for A .